Manu Nagendra Nalla

Software Engineer (Go, Distributed Systems, Kafka, Data Migration)
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PROFESSIONAL SUMMARY

Backend engineer with 6+ years building Go services, Kafka event pipelines and large-scale data migrations. At Kroger's Customer Platform: core engineer on a 1.22 billion-record production migration, 200+ Jira tickets completed, and creator of CP Krovis, a Kafka tool (Go, React, TypeScript) now used by 40+ developers across 3 to 4 teams. Holds an MS in Computer Science.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, JavaScript, SQL, Java, Bash
Backend and Messaging: Apache Kafka, Avro, Schema Registry, REST APIs, OpenAPI/Swagger, WebSockets, JWT, LDAP, Event-Driven Architecture, Dead Letter Queues
Databases: YugabyteDB, PostgreSQL, Redis, MongoDB, Cassandra, PL/SQL
Cloud and DevOps: Kubernetes, Docker, Rancher, Harness, Spinnaker, GitHub Actions, CI/CD, Azure, AWS, Akeyless, Disaster Recovery
Observability and Frontend: Dynatrace, Grafana, Structured Logging, React, Vite, Vitest
Practices: Agile, Scrum, Test-Driven Development, DevOps, Technical Design Documentation

PROFESSIONAL EXPERIENCE

Software Engineer II 05/2024 - Present
Ascendum Solutions (Client: Kroger), Cincinnati, OH

Customer Platform data migration and synchronization (Thesaurus team)

- Core engineer on the Customer Platform migration, delivering through every quarterly run from 24Q2 to close-out; owned the validation, error-analysis and fix-script streams for the final production run of about 1.22 billion records across 12 entity types, migrated at 99.99% accuracy with every residual delta on groups and alt IDs reconciled to zero. The platform reached 99% of EC and 81% of CG API traffic migrated by June 2026.
- Contributed to triage of about 8.1M errors from the April 2025 production run (CG and EC source data) and built a 14-category EC vs Netezza mismatch taxonomy that turned an open-ended reconciliation of about 200K records into fix-ready categories with named owners.
- Built a Go random-data generator that creates millions of synthetic records matching CG, EC and CP schemas; used in every migration run since 24Q3 and caught 15+ critical transformation bugs before production.
- Designed a reusable Go data-fix framework (fetchers for three source systems, analyzer, executor CLI, JSON Merge Patch library) that replaced one-off scripts, and used it to repair altID, householdID, shopperID, email and missing-card mismatches in production.
- Built a Failed Events API (dead-letter-queue-as-a-service) covering about 915K DLQ messages a month, and an error-handling package that records file, line and function at every error site across 22 files (2,755 lines), cutting debugging time by about 73% (2.5 hours to 40 minutes).
- Tuned forward-sync event consumption by load-testing transformer and loader counts against CPU allocation, raising throughput from about 4K to about 14K events per minute (3.5x) and finding that more workers add nothing once CPU is the limit (10 workers at 2000 mCPU peaked at about 14K; 20 workers fell to about 13K).
- Wrote code for the Kafka sync services that handle 60.6M production requests a month at a 0.0012% error rate and 7.8 ms p95, feeding Customer Platform back ends that serve 28.6 billion calls a month.
- Picked up 218 Jira tickets since May 2024 and completed over 200 of them across migration, validation, data fixes and design spikes; authored or co-owned 28 Confluence pages.
- Planned and executed seeding of the Harris Teeter indicator across about 10M CP Accounts records, delivered 29 production deployments from July to early September 2026 (9 data-fix releases) and moved 8 sync services to Akeyless external secrets.

CP Krovis: self-initiated Kafka explorer, built and owned end to end

- Initiated and built CP Krovis, now a production app used by 40+ developers across 3 to 4 engineering teams to browse, search, stream and publish Kafka messages without CLI access, replacing one-off consumers and platform tickets; referenced in the disaster recovery runbooks for finding Kafka offsets by timestamp during region failover.
- Unified 12 Kafka environments and 19 environment-region targets behind one interface, with Avro schema-registry decoding, multi-partition time-range search, live WebSocket streaming and JSON diff; 34 API endpoints, 23 React components, about 6,000 lines of Go.
- Secured it with LDAP, JWT sessions, rate limiting and audit logging; wrote 180 automated tests (115 Go, 65 Vitest) and set up CI/CD with GitHub Actions and Docker.

Customer Platform services (Phalcons team)

- Owned and released the Accounts, Groups, Search, Entities History, Orchestrator and Shim services: 14 production deployments across 25+ services from July to September 2026 under a formal release-governance process.
- Built disaster recovery for eight Kafka consumers and DLQ processors, including Central-to-East failover runbooks, and ran CP Search performance validation at about 2,995 GET requests per second (99.8% of target, 9.94 ms p95) and 585 POST requests per second (22.6 ms p95) against 150 ms and 300 ms SLAs.
- Operated production with Dynatrace, Grafana and Rancher and applied a layered test strategy of Postman, Karate automation across 5 repositories, Kafka consumer integration tests and Qmetry regression cycles.

Senior Software Development Engineer 04/2022 - 12/2022
ZopSmart, India

- Led a 10-member team replacing a legacy enterprise customer data system, increasing migration efficiency by 50%, and automated migration with custom scripts.
- Promoted from SDE I to Senior SDE in a double promotion after 2 years.
- Mentored junior developers on Go best practices, cutting their code review time by 50%.
- Gathered requirements across teams and delivered the Enterprise Customer project through the full SDLC in Agile.

Software Development Engineer I 03/2020 - 03/2022
ZopSmart, India

- Led an ETL-based migration project from inception to deployment; centralized customer data management, saving an estimated $2.3M annually.
- Increased real-time pipeline throughput by 60% with goroutines and channels, and cut server load by 25% with a Redis cache.
- Centralized PII data, avoiding CCPA penalties of up to $7,500 per incident.
- Built projects from scratch, created proofs of concept, set up CI/CD, and wrote unit and integration tests.

Application Development Associate 11/2019 - 03/2020
Accenture, India

- Supported Pfizer's PL/SQL systems: debugged stored procedures and triggers, optimized code for performance, and enhanced modules to meet business requirements.

PROJECTS

CP Krovis (2026): Internal Kafka message explorer built in Go, React and TypeScript and used by 40+ developers (details under Kroger experience).
recon (2026): Open-source Go CLI that reconciles two JSONL datasets and groups every difference by cause, with sample IDs, JSON output and CI-friendly exit codes (standard library only, with tests); github.com/manunag/recon.
Data Loader (2022): Go application that loads data between databases with configurable filters and transformations (CSV, JSON, SQL, Postgres, Cassandra, Kafka).

EDUCATION

Master of Science in Computer Science 01/2023 - 12/2025
California State University, Long Beach, Long Beach, CA
Bachelor of Engineering in Computer Science 07/2015 - 05/2019
Saveetha School of Engineering, Chennai, India
Relevant coursework: Advanced Algorithms, Advanced Artificial Intelligence, Machine Vision, Pattern Recognition.

CERTIFICATIONS AND ACHIEVEMENTS

- Certification: Anthropic, Claude Code in Action (February 2026), verify at verify.skilljar.com/c/nfpjfjyaymza.
- Instructional Assistant, Data Structures and Algorithms, CSU Long Beach; hosted tech sessions at ZopSmart; 1st place, Tame the Code, SAMVID, Saveetha University.